AI-Driven RF Design: An LLM Coding Agent Takes a Substrate Integrated Waveguide Filter from Specification to Fabrication

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***Abstract*—We report the design, fabrication and measurement of a third-order, 2.87 GHz substrate integrated waveguide bandpass filter. A large-language-model coding agent wrote and operated the entire toolchain: Chebyshev synthesis, parametric geometry, finite-element simulation with the open-source solver Palace, Bayesian optimization, and exporting fabrication files. One engineer set the specification, took the design decisions and ran the commercial-solver cross-checks. Three results stand out. First, a Gaussian-process optimizer applied to the assembled filter never reached a usable match. This is because the passband width it scores is discontinuous in the external coupling. Measuring smooth observables on small structures and inverting the coupling relations converged in about three solves per curve. Second, most of the 47 fixes in the project history corrected defects that produced plausible numbers without raising any error. We classify them and list the checks that now catch each class. Third, JLCPCB made the board on Rogers RO4350B with ENIG finish for about USD 120. We measured it with a vector network analyzer and recalibrated the data independently in software. It is centred at 2.894 GHz with 171 MHz bandwidth, 2.0 to 2.4 dB insertion loss and one broad reflection minimum near -18 dB. This is the floor of the calibration check. The simulation predicted 2.876 GHz, 182 MHz, 1.7 dB and three zeros. The measured loss matches an effective unloaded Q of about 140 to 185, depending on the estimator, as expected once copper loss is added to the simulation. The -16 dB specification is not met.**

***Index Terms*—Substrate integrated waveguide, bandpass filter, large language models, AI agents, design automation, Bayesian optimization, finite-element method, coupling matrix.**

Introduction

Machine learning has been part of microwave CAD for decades. Neural-network surrogate models [1], space mapping [2] and Bayesian optimization [3], [4] all replace some of the expensive electromagnetic (EM) solves with a cheaper learned or coarse model. Learned models are also used to extract coupling matrices from simulated responses, recently with physics-informed networks that need no labelled training set [5]. In each of these approaches the engineer still writes the geometry scripts, the meshing, the port setup, the objective and the export code. The engineer also interprets what the solver returns.

Recent large-language-model (LLM) agents work differently: they write and run code in a loop, observe the results and revise [6], [7]. Agent frameworks now exist for RFIC design [8] and RF amplifiers [9]. An agent has also operated CST, ADS and KiCad through their scripting interfaces to design a GNSS front end [10]. Empirical studies of bugs in agentic frameworks classify failures of the agent software itself [11]. These reports leave one thing largely unmeasured: how the *engineering* fails when an agent drives it. These failures are defects in the physical model that the agent writes. They matter more than the successes when one decides how much to trust the numbers.

This paper reports one complete design effort of that kind. The object is a classic structure, a three-cavity Chebyshev substrate integrated waveguide (SIW) filter [12], [13], [14]. We chose it because its physics is textbook, so every agent-produced number can be checked against theory. An LLM coding agent (Claude Code [15]) wrote and ran all of the design software. The author acted as specifier, reviewer and decision-maker. The project history is preserved in git, and most claims below map to a commit or a stored result. Some process statistics survive only in engineering notes written at the time, not in raw logs. One example is the screen-out rate of the Bayesian search. We flag these where they appear.

The contributions are:

1. An account of the tool path. A finite-difference time-domain (FDTD) solver (openEMS) could not support match tuning, but a finite-element solver (Palace) could. A GPU workstation, run through remote agent sessions, had memory limits that eventually pushed all production runs onto CPUs (Sec. III).
2. A measured explanation of why black-box Gaussian-process (GP) search over the assembled filter fails, and of why calibrate-and-invert succeeds (Sec. IV).
3. A taxonomy of the silent modelling defects the process produced and caught, with the guard that now blocks each class (Sec. V).
4. The fabricated design, and a diagnosis of its deviation from the specification by coupling-matrix

extraction. This item also covers a documented dual-mode negative result, an insertion-loss budget, a launch study and the manufacturing path. It ends with the measurement of the fabricated board with a vector network analyzer (VNA), and the quality factors read from it (Secs. VI to IX).

Filter specification and synthesis

The specification is f0 = 2.87 GHz, 5 % fractional bandwidth (FBW, 143.5 MHz), 0.1 dB Chebyshev ripple and $|S11| \le -16$ dB in band. The laminate is Rogers RO4350B, 0.76 mm thick ($\varepsilon_r = 3.66$, $\tan\delta = 0.0037$ [16]), with 0.5 oz copper. The fence vias are 1.5 mm in diameter at a 3.0 mm pitch. Their p/d = 2 ratio is at the edge of the leakage rules of [17], [18]. Smaller vias (0.8 mm at 1.5 mm pitch) were compared three times under three different objectives and never matched this choice. The connector is a Taoglas EMPCB.SMAFSTJ.B.HT edge-mount SMA [19].

Design equations

The low-pass Chebyshev prototype gives $g_1 = g_3 = 1.0316$ and $g_2 = 1.1474$ for $N = 3$ and 0.1 dB ripple [20]. The bandpass mapping turns them into two numbers that the geometry must reproduce: the coupling between adjacent cavities and the external quality factor of the end cavities:

$$k_{i,i+1} = \frac{\mathrm{FBW}}{\sqrt{g_i\, g_{i+1}}} = 0.0460, \qquad Q_e = \frac{g_0\, g_1}{\mathrm{FBW}} = 20.63. \quad (1)$$

We choose the TE10 cutoff near $0.7 f_0$, which sets the equivalent width. The Cassivi–Wu correction then gives the real width for the finite via fence [21]:

$$a_{\mathrm{eff}} = \frac{c}{2 f_c \sqrt{\varepsilon_r}}, \quad f_c \approx 0.7 f_0, \qquad a_{\mathrm{siw}} = a_{\mathrm{eff}} + \frac{d^2}{0.95\, p}. \quad (2)$$

Each cavity length follows from the TE101 resonance condition:

$$f_{101} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\frac{1}{a_{\mathrm{eff}}^2} + \frac{1}{L^2}}. \quad (3)$$

Eq. (3) assumes ideal walls, so the real length is shorter once the tapers and apertures load the cavity (Sec. VI). The feed is a 50 Ω microstrip line (Hammerstad [22]) flared into the SIW by a Deslandes taper [23]. At the board edge it becomes a grounded coplanar waveguide (CPWG) launch sized to the SMA footprint. The launch has a 1.2 mm trace with 0.2512 mm gaps. Conformal mapping gives 50.0 Ω for these dimensions [24], [25].

The two couplings are measured, not computed, because closed-form iris formulas are unreliable for SIW. Two coupled closed cavities split their resonance into f_1 and f_2, and a single cavity driven through its apertures has a loaded quality factor Q_L:

$$k = \frac{f_2^2 - f_1^2}{f_2^2 + f_1^2}, \qquad Q_e = \frac{2}{1/Q_L - 1/Q_u}. \quad (4)$$

The coupling-matrix formulation [26], [27] provides the ideal target response and, later, the diagnostic fit. The unloaded quality factor Q_u sets the dissipation of a narrowband filter [20]:

$$\mathrm{IL}_0 \approx \frac{4.343 \sum_i g_i}{\mathrm{FBW}\, Q_u}\ \mathrm{dB}. \quad (5)$$

Toolchain and working method

Fig. 1 shows the flow and the division of labour. Fig. 2 shows how the effort was spread over time.

Phase 1: analytic design, openEMS and CST

The first iteration (June 2026) used analytic dimensions. It used the free FDTD solver openEMS [28] for calibration and CST Studio Suite 2021 [29] as the reference. Python generated a parametric CST VBA macro. The macro builds native solids wrapped in the solver's history list. The analytic cavity length of 38.23 mm put the CST passband at 2.818 GHz with a 293 MHz band, 3.65 dB insertion loss and only 1.02 dB return loss. The external coupling was uncalibrated and heavily over-coupled.

openEMS was abandoned for three reasons. First, FDTD resolves the 1.5 mm cylindrical vias on a staircase grid. This needs at least three cells per diameter over the whole board. The resonant, high-Q structure also rings down slowly, so each full-filter run took minutes to tens of minutes. Second, the ports did not give a trustworthy S11. The lumped port radiated like a point source instead of launching the microstrip mode. The microstrip port launched the mode correctly, but its S11 de-embedding was unstable and even returned $|S11| > 1$. The passband and S21 were reproduced, but the match could not be tuned. Matching was the whole remaining problem. Third, openEMS had no eigenmode solver for the coupling calibration of Sec. IV. The calibration had to be emulated by reading peak splits off driven sweeps.

Palace [30] removed all three obstacles. It is an MPI-parallel finite-element solver built on MFEM [31] with gmsh meshes [32]. Its tetrahedral mesh conforms to the vias. Its wave ports solve the 2D port mode, as CST's do, and are passive by construction. It has an eigenmode solver. JSON files drive it entirely, which makes it easy for an agent to operate. A first Palace model reproduced the CST result for the same geometry and then replaced openEMS in the loop.

Phase 2 onward: an agent-operated toolchain

From August the agent consolidated three copied optimizer scripts into one package with a `Topology` interface and a single command-line tool (`siw`). The tool covers synthesis, layout, eigenmode and driven solves, Bayesian

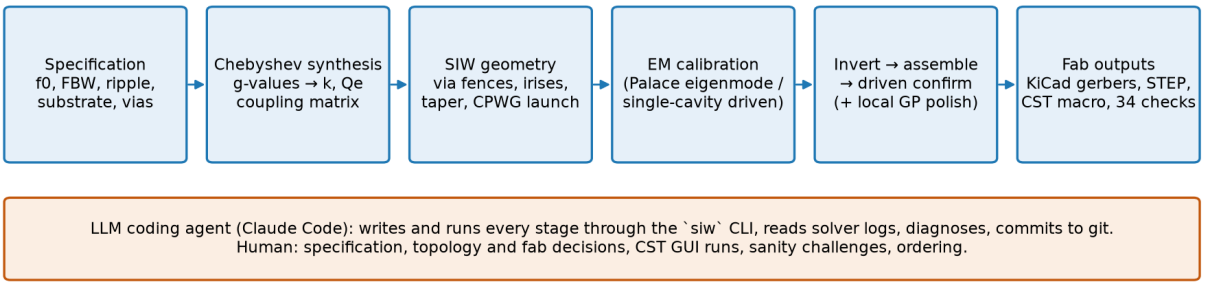


Fig. 1. Design flow. The agent writes and operates every stage through one command-line tool (`siw`). The engineer supplies the specification and the decisions, and runs the commercial-solver cross-checks.

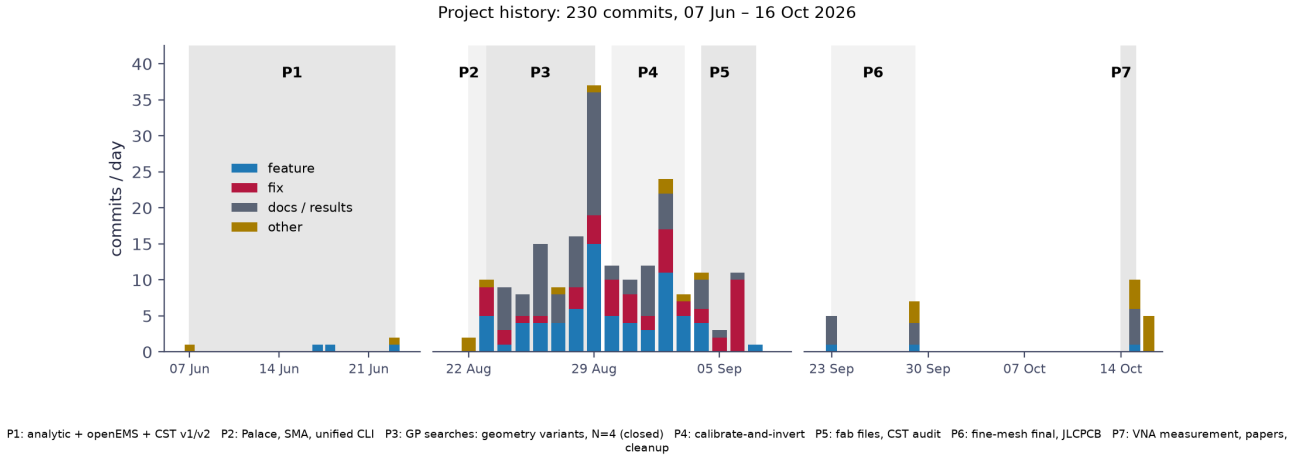


Fig. 2. Commit history by type. P1 was a human-led prototype committed in a few large commits. From P2 onward the agent drove the work, and fixes (red) cluster where new physics or a new tool entered the loop.

design, 1-D scans, calibration, comparison figures and CAD export. Every production number in this paper can be reproduced with that tool. Curated results are stored in a `golden/` directory together with their S-parameters and the command that reproduces them. A project brief file (`CLAUDE.md`) acted as the agent's persistent memory across sessions. Handoff documents carried the state from one machine to another.

From the textbook steps to the tool. Table I lists the steps that a coupled-resonator SIW filter normally takes by hand and the command that replaces each step. The equations are the classical ones [18], [20], [21]. The tool changes two things. Every step that used to be a hand-fitted curve is now a measurement with a solver. The chain from specification to gerbers is now one sequence of commands.

The optimizer. The search code is our own, written around Palace, not a package taken from a design tool. A candidate is a set of geometry parameters: the cavity lengths (mirrored about the centre), the iris widths, the external aperture and the taper length. The microstrip width stays at its 50 Ω value. To evaluate a candidate, the code meshes the whole board with gmsh, including the connectors and the launch. It then writes the Palace configuration and runs a driven sweep on a coarse mesh. Finally it reads $|S_{11}|$ and $|S_{21}|$. A fast physics screen at f_0 runs first. It rejects candidates that do not resonate. The rejected ones get a graded penalty, so the surrogate still sees a gradient. The objective sums these terms. The first is the root-mean-square error (RMSE) of $|S_{11}|$ against the ideal Chebyshev response (both clipped at a floor). The others are the shape of $|S_{21}|$, the insertion loss, the centre-frequency error outside a deadband, a two-sided bandwidth error and the board length. Three one-sided hinges add the real specification: worst in-band $S_{11} \leq -16$ dB, loss within an allowance over the dielectric floor, and symmetry. The objective went through ten revisions (v9–v18). The early ones scored individual points, and the later ones fitted the whole response. A Gaussian-process surrogate (scikit-optimize [33], [34], `gp_hedge` acquisition) proposes each next candidate. A random seeding phase comes first, and its size scales with the budget. A patience rule stops the search early on a plateau. The winner is solved again on a fine mesh, and the result is checked after every dimension is rounded to the 0.1 mm fabrication

TABLE I
THE DESIGN STEPS OF AN SIW COUPLED-RESONATOR FILTER: THE TEXTBOOK WAY AND WHAT THE TOOL DOES.

	Step	By hand	In the tool
1	Synthesis	Chebyshev g_i by recursion; $k_{i,i+1} = \mathrm{FBW}/\sqrt{g_i g_{i+1}}$, $Q_e = g_0 g_1/\mathrm{FBW}$; ideal response from the coupling matrix	`synthesis.py`: targets k, Q_e and the ideal `target.s2p`
2	Waveguide width	$a_{\mathrm{eff}} = c/(2f_c\sqrt{\varepsilon_r})$ with $f_c \approx 0.7 f_0$; $a_{\mathrm{siw}} = a_{\mathrm{eff}} + d^2/(0.95p)$; via rules on p/d	`siw_geometry.py`, from the substrate and via presets in `config.py`; builds the via fences and the board
3	Cavity length	$f_{101} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{a_{\mathrm{eff}}^{-2} + L^{-2}}$	first guess only; corrected from a driven solve in step 7
4	Feed	$50\,\Omega$ microstrip width (Hammerstad), taper (Deslandes), coplanar gap by conformal mapping	`siw_geometry.py`, `launch.py`: taper and the CPWG footprint of the SMA
5	Internal coupling	simulate two coupled cavities, read the split $k = (f_2^2 - f_1^2)/(f_2^2 + f_1^2)$, fit $w(k)$	`siw calibrate --what k`: Palace eigenmode of a closed two-cavity model, polynomial fit, inverted at the target k
6	External coupling	one resonator, Q_L from the -3 dB width or the group delay, $Q_e = 2/(1/Q_L - 1/Q_u)$	`siw calibrate --what qe`: one driven cavity with two apertures, fit of $w_{\mathrm{ext}}(Q_e)$
7	Assembly and tuning	simulate the whole filter, change lengths and irises by hand until the reflection zeros are equiripple	`siw synth-design`: joint (k, f_c) refinement, driven centre-frequency trim, separate middle-cavity correction, pole-shape check
8	Final polish	manual sweeps of one parameter at a time	`siw design --seed`: local Bayesian search (below)
9	Layout and CAD	CST model drawn by hand; gerbers from a PCB tool	`siw cst`, `siw kicad` and 34 file-level checks
10	Verification	VNA measurement and calibration	`siw measure`: software recalibration and comparison with the simulations

resolution. In local mode (`--seed`, `--span`) the search box is a few millimetres around a calibrated design. A coarse evaluation takes 6–20 min, so sample efficiency matters more than throughput. A lock serialises the Palace solves.

Compute. To speed up the search, we moved to a workstation with an NVIDIA GTX 1660 SUPER (6 GB) and ran Palace there with its GPU backend. The author supervised the long agent sessions remotely through Claude Code's Remote Control. On the dual-mode model the GPU was 1.24× faster than 4 CPU ranks. The S-parameters were identical to 10 dB. Disabling MFEM's pooled allocator (`MFEM_MEMORY=cuda`) removed the first out-of-memory failures. Those failures had killed every point of an eigenmode sweep. But the card does not have enough memory for the meshes that matter. It finished a driven solve with 174.6 k tetrahedra and ran out at 178.4 k, in the error estimator. Only one rank per card works. The 166 mm cascade exceeds that limit at any adequate mesh density, so no GPU mesh preset was ever validated for it. The code now caps GPU use at 176 k tetrahedra and falls back to the CPU. We did not try the GPU on the fine-mesh verification. That verification needs 315 k unknowns and about 5.5 GB on the CPU. All production runs therefore ran on CPUs with 4 MPI ranks. Four attempts on a 3.7 GB host failed, three of them explicitly out of memory. The final runs used a 16 GB host at about 8 s per frequency point. Memory, not arithmetic, was the binding constraint. A card with more memory would bring the cascade onto the GPU. So would the half-model and adaptive sweeps of Sec. X.

Who did what: models and effort

The commit trailers and the agent's session logs record which model was working and for how long. Table II and Fig. 3 summarise them. They come from `docs/paper/make_process_stats.py`, which keeps only counts, with no prompts and no file contents.

What the numbers show:

- *Two models worked side by side.* Sonnet 5 and Opus 5 were both active from 22 August to early September, with about the same share of commits (64 and 43) and fixes (16 and 14). Sonnet 5 produced 70 % of the output tokens of that period. The logs record the model of every turn but not why it was chosen. The later work used Opus 5.5 (23–29 September, the fabrication files and the final checks) and Sonnet 5.5 (15 October, the measurement and this text). One 48-turn run on Fable 5 on 28 August produced one commit.
- *Sub-agents did a small part.* There were 27 sub-agent runs, 26 on Sonnet models and one on Opus 5. They account for 4 % of the output tokens.
- *Effort was concentrated.* The agent was active for 135 hours on 21 days. The busiest days, 24, 28 and 31 August, each had 14–15 hours. The engineer sent about 500 messages, which is 24 per active day.
- *Most of the work was running things.* The logs hold 4,393 shell commands (3,816 in the main thread and 577 in sub-agents), 738 file edits, 758 file reads and 74 web searches. The agent also called a timer 874 times and a monitor 214 times. These calls waited for solver runs that lasted from minutes to hours, without holding the conversation.
- *Fixes were a fifth of the history.* 47 of 220 commits are fixes (21 %). Of the 125 commits with a trailer, 30 are fixes. The trailer is a lower bound for agent work. Of the 198 commits between 22 August and 10

TABLE II

MODELS USED, FROM THE SESSION LOGS AND THE COMMIT TRAILERS. TURNS ARE ASSISTANT MESSAGES IN THE MAIN THREAD, PLUS THOSE OF SUB-AGENTS. OUTPUT TOKENS COVER BOTH. FIX COMMITS ARE THOSE WHOSE MESSAGE STARTS WITH `FIX`. COMMIT COUNTS ARE FOR THE HISTORY UP TO COMMIT `4D570AC`.

Model	Main-thread dates (2026)	Turns (main + sub-agent)	Output tokens (M)	Commits with trailer	of which fix
Sonnet 5	22 Aug–10 Sep	9 287 + 1 362	8.41	64	16
Opus 5	22 Aug–4 Sep	3 673 + 65	3.57	43	14
Fable 5	28 Aug	48	0.06	1	0
Opus 5.5	23–29 Sep	922	0.73	12	0
Sonnet 5.5	15 Oct	424 + 200	0.63	5	0
Total		14 354 + 1 627	13.39	125	30

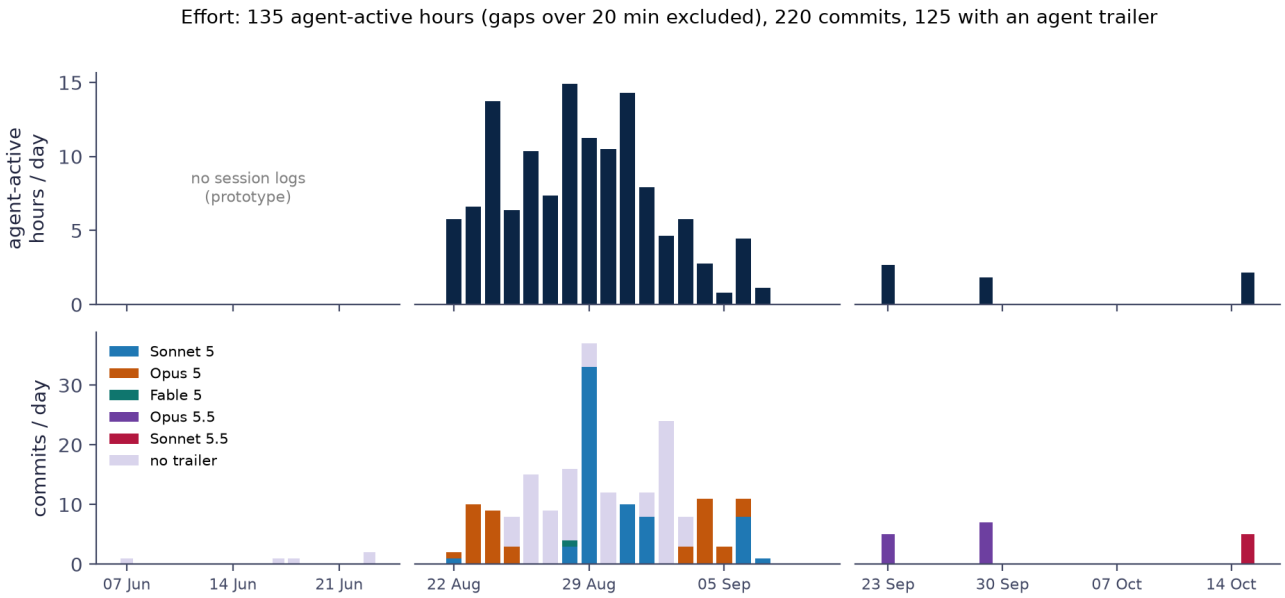


Fig. 3. Effort over time. Top: hours per day in which the agent was active, counting only gaps shorter than 20 min between logged events. Bottom: commits per day, coloured by the model named in the commit trailer. Grey commits have no trailer.

September, 90 have no trailer. The logs show the agent working on every one of those days.

- *Code was rewritten.* The history adds 16.1 k lines of Python and removes 5.9 k, including the clean-up of 15 October. A typical commit touches two files. The package now has 12.4 k lines.

WHY GLOBAL BAYESIAN SEARCH FAILED

The GP worked as intended in one regime: as a local polish around a validated design. For the dual-mode filter, a 30-evaluation search in a ±2 mm, 5-dimensional box confirmed the landed geometry. Neighbouring probes came back worse, and the apparent 0.41 dB gain lay inside the measured coarse-mesh noise floor of 0.64 dB.

Run globally over the assembled cascade, the same machinery never produced a worst in-band S11 better than −5 dB in more than 60 driven evaluations. The cause is structural. The objective scores the passband width. That width is discontinuous in the external aperture w_{ext}. Below a coupling threshold, only the centre pole clears the −3 dB line, so the band measures one resonance. Above the threshold, all N poles merge and the width jumps. On the N = 3 cascade the widths recorded during the search were 16, 22, 36, 65 and then 215 MHz. The 143.5 MHz target lies inside the jump. We re-measured this for the paper on the fabricated geometry (Fig. 4c, data in the repository). The width rises 96, 106, 123 MHz for w_{ext} = 10–12 mm and jumps to 195 MHz at 13 mm. It then *falls* again, because the end cavities become over-coupled and the reflection zeros drop from three to one. The target width is crossed twice. The second crossing, at 16 mm, has one reflection zero and −4 dB of return loss. A bandwidth-driven search lands there, and so does the scan's own inversion (15.2 mm). A GP assumes a smooth objective, so it inherits both the jump and the false solution. The search-health data showed the same thing. On the cascade, 57–83 % of candidates were screened out, which left 2–9 informative points per run. On the dual-mode filter, where the search converged, 34–37 % were screened out and there were about 19 informative points.

The replacement measures each coupling on the smallest structure that shows it (Fig. 4):

- **Internal coupling.** A closed two-cavity eigenmode solve splits the cavity mode into f_1 and f_2, and k follows from Eq. (4). It is smooth in w_{iris}.
- **External Q.** One cavity driven symmetrically

through two apertures gives the loaded Q, and Q_e follows from Eq. (4). Here $Q_u = 1/\tan\delta = 270.3$. The eigenmode Q independently matches this value. It is smooth and monotonic in w_{ext}.

We inverted the fitted curves at the synthesis targets, assembled the filter and made a final driven trim of the centre frequency (fc). This converged in about three solves per curve. The re-run calibration for this paper inverts to $w_{\text{iris}} = 13.85$ mm and $w_{\text{ext}} = 13.53$ mm. The fabricated board has 14.42 and 13.96 mm, and its end cavities were trimmed afterwards. The resulting internal coupling was within 2 % of target. At that checkpoint the N = 4 centre frequency also landed at 2.870 GHz. The N = 4 design was later abandoned. Its best return loss was −6.1 dB (Sec. VI). The GP was kept only as a local polish seeded with the result (`--seed`). The physics that worked is textbook [20], [27]. The agent's contribution was to run the calibrations quickly and to diagnose why the black-box route had failed.

Two further search failures came from the parameterization itself, not from the search. The via fences were first generated by filtering a fixed grid, so the iris width was *quantized*. The whole cascade search box spanned about two distinct geometries. The GP was tuning a knob that did nothing (Fig. 5). In addition, N ≥ 4 had a single shared iris width and could not express the required k = [0.0415, 0.0329, 0.0415].

Silent defects: a taxonomy

As of 15 Oct 2026 (up to commit `4d570ac`) the history holds 220 commits, 125 of them with an agent co-author trailer, and 47 labelled *fix*. Most of the fixes corrected defects that never raised an error. The solver converged and the plot looked reasonable, but the number was wrong. Table III groups representative cases. Three rules paid off repeatedly. First, if a fix does not change the number, the diagnosis was wrong. Second, after two wrong inferences, stop inferring and run a one-factor ladder. Third, check every result against what physics allows: insertion loss cannot beat a lossless model, the pole count must equal N, and a lossy stack cannot cost 27 dB.

Two cases changed the course of the project. In the first, a real SMA coax launch replaced the ideal port. The optimized microstrip width of 2.26 mm then turned out to be a 41 Ω line. The optimizer had exploited a port that normalizes to whatever it sits on. The true 50 Ω width is 1.664 mm. The first coax model also used a full-size 0.82 mm pin, which is thicker than the 0.76 mm board. It returned −0.2 dB S11. That was real physics, not a bug (Fig. 6). In the second, an independent audit against CST showed that the −13.37 dB return loss was a coarse-mesh artefact. We had used that value to choose the cascade over the dual-mode design. The converged value is about −10.7 dB. The decision survived, but the justification had to be rewritten.

Topologies, lengths and the dual-mode negative result

Cavity lengths. The analytic 38.23 mm cavity placed the band at 2.82–2.905 GHz (openEMS/CST). An openEMS sweep to 35.93 mm and 33.64 mm moved it to 2.982 and 3.077 GHz, a slope of about −37 MHz/mm. Palace models the loading by the tapers and external apertures, which the analytic formula ignores. With Palace the cavities shortened to about 34.4 mm. Calibrate-and-invert then gave unequal cavities, 34.24/35.50/34.24 mm. The fine-mesh coupling fit found the end cavities about 17 MHz low, so a −0.3 mm trim (33.94/35.50/33.94 mm) was chosen. A −0.6 mm trim gained only 0.07 dB and cost 0.4 dB of band-edge loss. The measured sensitivity at the end cavities was about 27 MHz/mm, roughly half the 50 MHz/mm rule of thumb.

Board size. We also explored the design space for a smaller board (Figs. 7 and 8). The candidates were a single-cavity dual-mode N = 2 filter (119 mm), the N = 3 cascade (166 mm) and an N = 4 cascade (211 mm). We also tried three other single-cavity shapes: corner-cut, circular-cross and rhombic. None came closer to the specification than the dual-mode cavity (best S11 −6 to −9 dB). We closed the compact N = 4 made from two dual-mode cavities, after five mechanisms and a joint GP search all stopped at two of four poles. The five mechanisms were offset apertures, near-corner coupling, top-copper slots, discrete posts and quad-mode cavities. A four-cavity single-mode cascade works, but it needs 180–206 mm.

Dual-mode N = 2. A single square SIW cavity, a = 57.16 mm, carries a degenerate TE102/TE201 pair. A floating internal via row splits the pair and couples it [35], [36]. Under the matched CPWG launch its best response is fc 2.904 GHz, 316.6 MHz bandwidth, −10.76 dB return loss and 1.45 dB insertion loss. Real copper adds 0.32 dB. One-dimensional scans on the assembled filter show that bandwidth and match are *opposed* on every geometry parameter of this topology. At $w_{\text{iris}} = 11$ mm the band is 145.6 MHz (4.94 % FBW), but the return loss is −1.75 dB. The pad-only launch used earlier reached −17.7 dB. That launch was not the one fabricated. N = 2 has only one internal and one external coupling. This is not enough freedom to place two poles and also match them. The cascade has an extra degree of freedom at 1.4× the length, and it won the head-to-head comparison.

The fabricated design

The chosen geometry is $l_{\text{cav1}} = 33.939$, $l_{\text{cav2}} = 35.503$, $w_{\text{iris}} = 14.416$, $w_{\text{ext}} = 13.96$, $l_{\text{taper}} = 15.738$ and $w_{\text{ms}} = 1.6636$ mm. The board is 165.78 × 45.79 mm. Fig. 9 shows the fine-mesh result (2.5/0.45 mm volume mesh, 0.15 mm coax ports, 9.12 mm air, 61 points). The result is fc 2.876 GHz, −3 dB bandwidth 182.3 MHz and worst in-band $|S_{11}|$ −11.15 dB. Insertion loss is 1.70 dB at the peak and 2.58 dB at the band edge. Table VI (Sec. VIII) sets it against the specification, CST and the measurements.

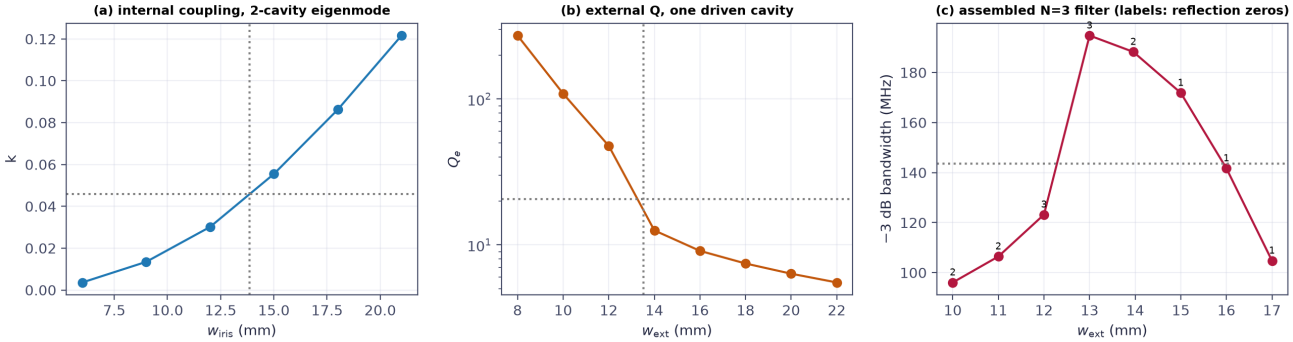


Fig. 4. (a), (b) The calibration observables are monotonic and invertible (dotted: synthesis target and inverted width). (c) The −3 dB bandwidth of the assembled filter, the quantity a black-box objective scores, jumps between 12 and 13 mm and then falls. Labels: number of reflection zeros. Coarse mesh.

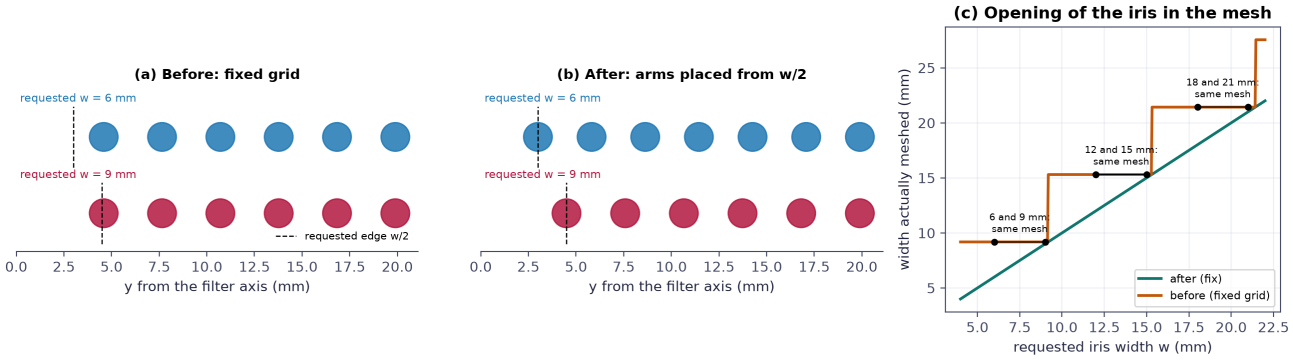


Fig. 5. Via-fence quantization. (a) With the fixed grid, requesting an iris of 6 mm or 9 mm placed exactly the same vias (dashed: requested edge, $w/2$). (b) With the fix the iris arms start at $w/2$ and every requested width gives a different mesh. (c) Opening actually meshed versus requested width: steps of about 6 mm before, a straight line after. Six requested widths fell on only three geometries.

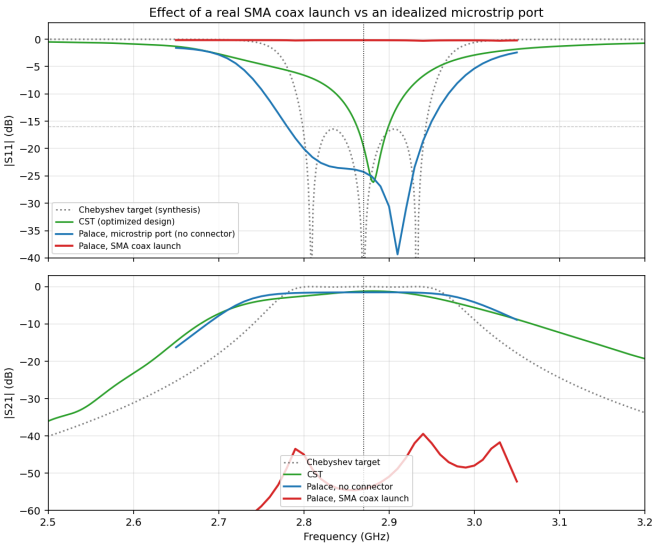


Fig. 6. First real-connector model. An ideal port hid a 41 Ω feed, and a cable-size pin larger than the board shorts the launch.

Why it misses the specification. A lossy N = 3 coupling-matrix fit to the refined sweep of the untrimmed board (Fig. 10) puts the external coupling on target (+0.4 %), which exonerates the launch. It finds the internal coupling 4.8 % weak and the end cavities 17 MHz low. The trim reduces the detuning. The same fit on the fabricated geometry gives the end cavities 11 MHz low. It gives the internal coupling 8 % weak and the external coupling 4.5 % weak (Q_e 22.6 against 20.6). The worst in-band point sits at the upper band edge. There the equiripple region ends near 2.935 GHz, against a target edge of 2.942 GHz. Adding +0.6 mm of w_{iris} on top of the trim widened the band without helping. Adding +0.4 mm of w_{ext} on top of both was strongly harmful, and also pulled the end cavities down. No isolated w_{ext} trial was run. A coordinated local move remains open.

Is the insertion loss (IL) what we should expect? Yes, to within the measurement uncertainty. A hand calculation shows this. It needs only a few numbers. It also shows how much the real cavity length and the real copper change the answer.

Step 1: the loss formula. The unloaded Q, Q_u, is the quality factor of one cavity on its own, without the external circuit. It says how much energy the cavity stores compared with what it loses in each cycle, so a higher Q_u means a lower insertion loss. It sets the loss of a narrowband filter, Eq. (5). For $N = 3$ the sum of the prototype elements is $\sum g_i = 3.21$, and the design bandwidth is FBW = 0.05, so

TABLE III
REPRESENTATIVE SILENT DEFECTS, HOW THEY WERE CAUGHT, AND THE GUARD ADDED

Class	Defect (effect)	Caught by	Guard now in code
Parameterization	Iris width quantized by grid filtering (search box ≈2 geometries); single iris width for $N \geq 4$ (spec unreachable); stale via defaults 0.8/1.5 mm	Scan showing identical meshes; coupling targets	Continuous via segments; per-iris parameters
Measurement / metric	Bandwidth taken first-to-last crossing, bridging two disjoint lobes; sweep window narrower than the band (fc reported = window centre, 48 MHz error charged nothing); truncated Q_e points giving an inversion outside its own bracket	Identical fc across evaluations; physics sanity	`n_lobes`, `band_truncated`, bracket check
Mode selection	Eigenmode pair = smallest gap picked a higher-order cluster: $k = 0.0028$ instead of 0.0583 (21×)	Inspecting the raw eigenvalue list	Window around f_0, then the two lowest modes
Mesh / boundary tagging	SMA end-cap classified by centroid swallowed the PTFE annulus at $r_{in} \geq 0.3773$ mm (S21 = −79.5 dB); faces left with no boundary condition (IL better than lossless); a domain with no material	Bisection over pin radius; IL > lossless	Face-extent test; unclassified-face report; domain-coverage guard
Model validity	Wave port on the trace self-normalizes, so W_{ms} was tuned to a 41 Ω line; surface-impedance BC applied to internal two-sided sheets (−7.8 dB IL on copper); 4.5–20 μm layers meshed as slivers	Real coax launch; Palace warning; one-factor ladder	Real SMA model; exterior-only conductivity; thin layers excluded
Plumbing	Adapter dropped soldermask/via-conductivity before the solver (two studies ran with the feature off); N=3 and N=4 results overwrote each other (shared tag); cached curve reused under a retuned label	Result unchanged by a real change	Order in every tag; geometry fingerprint on caches
Export	CST macro wrote the launch at 1000× (6 m coax, vias 30 m off-board), twice; taper dropped; coax without shield; PEC end caps on open walls	Cross-check of `.bas` vs mesh	34 file-level fab checks incl. magnitude test
Agent process	VBA properties guessed by name analogy; edits from a parallel agent absorbed unreviewed; comment overclaims; a process declared dead by a flawed `ps` filter	CST errors; adversarial self-review	Only properties seen in a real run; reviewed diffs
Convergence	Selection metric (S11 −13.37 dB) from a non-converged mesh; refined model gives −10.7 dB	CST/Palace audit	Fine-mesh final check

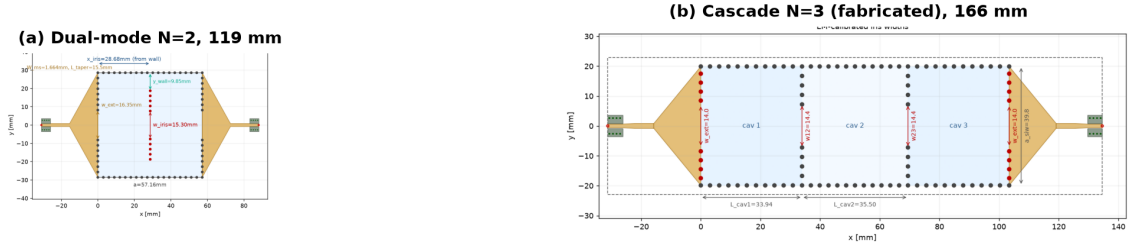


Fig. 7. (a) Dual-mode N=2: one square cavity whose TE_{102}/TE_{201} pair is split by a floating via row. (b) The fabricated N=3 cascade, with external apertures in red and the CPWG/SMA footprints at the board edges.

$$\mathrm{IL}_0 \approx \frac{4.343 \times 3.21}{0.05\, Q_u} = \frac{279}{Q_u}\ \mathrm{dB}. \tag{6}$$

Step 2: the Q of one cavity. The dielectric and the copper act in series,

$$\frac{1}{Q_u} = \frac{1}{Q_d} + \frac{1}{Q_c}, \qquad Q_d = \frac{1}{\tan\delta} = 270. \tag{7}$$

The dielectric alone gives 279/270 = 1.03 dB. For the copper we use the conductor Q of a TE_{101} cavity [37],

$$Q_c = \frac{(kaL)^3\, b\, \eta}{2\pi^2 R_s} \, \frac{1}{2a^3 b + 2bL^3 + a^3 L + aL^3}, \tag{8}$$

with $k = 2\pi f\sqrt{\varepsilon_r}/c$ and $\eta = 377/\sqrt{\varepsilon_r}$. The effective width is $a = 39.0$ mm (Sec. II) and the substrate height is $b = 0.76$ mm. The cavity length is L. The surface resistance of copper at 2.87 GHz is $R_s = 14$ mΩ (skin depth $\delta = 1.23$ μm). The fabricated end and middle cavities, 33.9 mm and 35.5 mm long, give $Q_c = 519$ and 547, so Q_c is about 530–540 and we use 540. Copper alone then costs 279/540 = 0.52 dB. The two together give $Q_u = 180$ and 1.55 dB.

Step 3: from the cavities to the board. Palace models the walls as perfect conductors, so its 1.70 dB contains the dielectric but no copper. It is 0.67 dB above the 1.03 dB of the formula. The simulated board also has about 60 mm of 50 Ω feed line, the SMA transition, radiation from the open microstrip sections and the mismatch between the

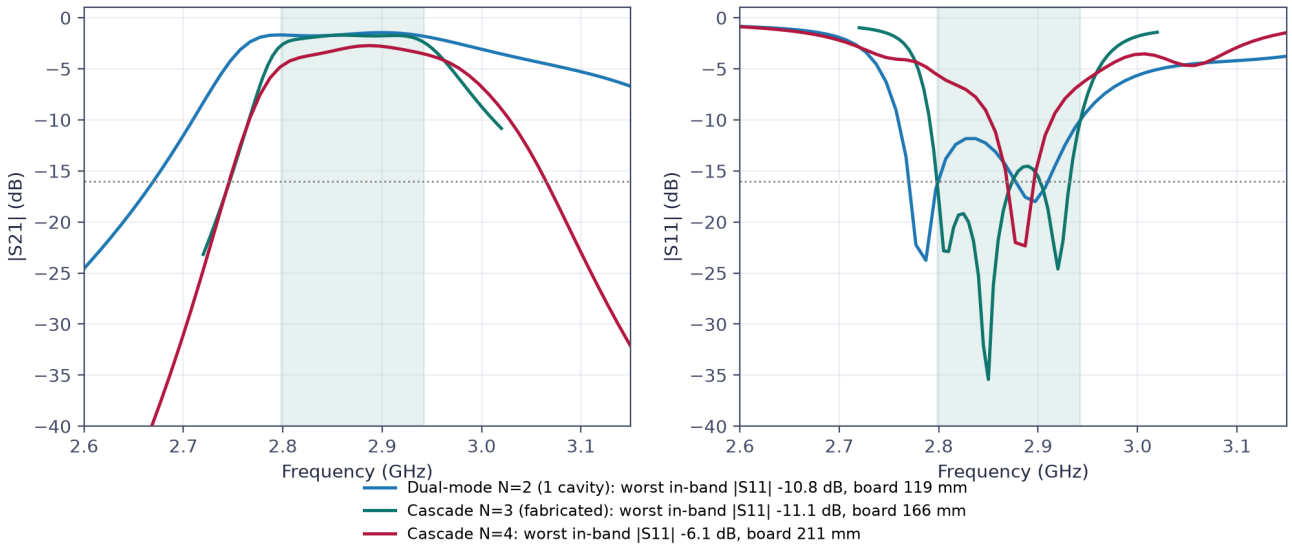


Fig. 8. Best designs of each topology under the same CPWG + SMA launch (Palace). The N=3 curve is the fine-mesh final. The other two are the production mesh.

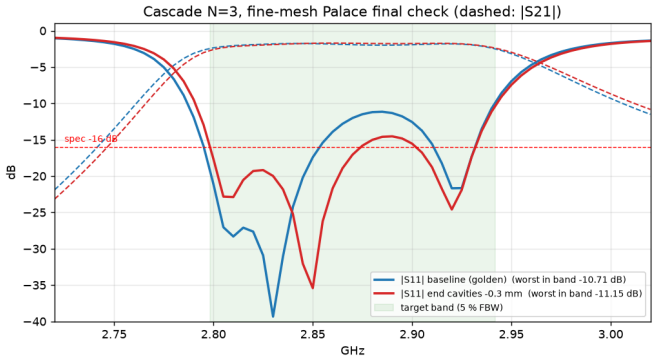


Fig. 9. Fine-mesh final check. The −0.3 mm end-cavity trim lifts the interior ripple from −11.2 to −14.5 dB. The worst point stays at the 2.94 GHz band edge.

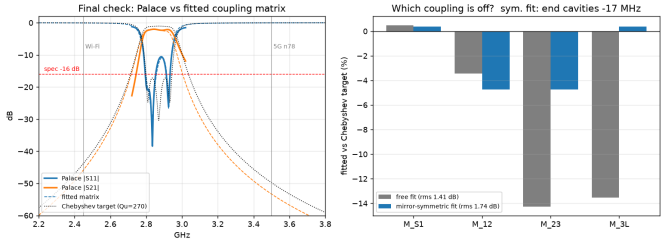


Fig. 10. Lossy coupling-matrix fit used to diagnose the return-loss gap. Untrimmed board: external Q on target, k 4.8 % weak, end cavities 17 MHz low.

reflection zeros. The formula contains none of them. We did not separate these terms. Only the feed line is easy to estimate. On this laminate a 50 Ω line loses 1.5 dB/m in the dielectric and 1.5 dB/m in smooth copper. So 60 mm cost 0.09 dB in each.

Step 4: add the copper to the simulation. The copper is an extra term, so the expected loss is 1.70 + 0.52 + 0.09 = 2.3 dB (Table IV). Real foil is rough. The Hammerstad factor $K = 1 + \frac{2}{\pi}\arctan(1.4(\Delta/\delta)^2)$ multiplies the conductor loss. For an rms roughness Δ of 0.5, 1 and 2 μm it is 1.14, 1.47 and 1.83. The total then becomes 2.4, 2.6 and 2.8 dB. The measured 2.0–2.4 dB (±0.3 dB) is at or below the smooth-copper value, so there is no unexplained loss. The data cannot tell smooth copper from a roughness of up to about 1 μm rms.

TABLE IV
INSERTION-LOSS BUDGET AT 2.87 GHz (dB), FROM EQS. (6)–(8).

Cavity dielectric ($Q_d = 270$)	1.03
Feed lines, SMA launch, radiation, mismatch (rest of the Palace result)	0.67
Palace, perfect-conductor walls	**1.70**
Copper in the cavities ($Q_c \approx 540$)	+0.52
Copper in the feed lines (about 60 mm)	+0.09
Expected, smooth copper	**2.3**
Same, rms roughness 0.5 / 1 / 2 μm	2.4 / 2.6 / 2.8
Measured (Sec. VIII)	**2.0–2.4**

The first version of this budget was written before the board was measured and gave 2.1–2.3 dB. It used the simulated 6.3 % bandwidth in the loss formula instead of the 5 % of the prototype, which the g_i belong to. It also used a copper term of 0.4 dB. With the corrected terms, the dielectric part moves from 0.8 to 1.0 dB, the copper part from 0.4 to 0.5 dB, and the total from 2.2 to 2.3 dB. The 0.52 dB for copper agrees with two other estimates. The coupling-matrix model adds 0.55 dB when Q_u falls from 270 to 180. An exterior-face copper model in Palace adds 0.59 dB. This is a lower bound, because the internal trace and ground sheets stay ideal there.

How the real length and the copper move the answer.

- *Cavity length.* Q_c grows with the size of the cavity: 519 at 33.9 mm, 593 at the analytic 38.2 mm of Phase 1. The trimmed cavities therefore lose about 0.07 dB more than the analytic ones would. A length tolerance of ± 0.1 mm moves the centre frequency by 2.7 MHz (27 MHz/mm) but Q_c by only 0.3 %, so it does not change the loss.
- *Board length.* Each 10 mm of 50 Ω feed line costs 0.03 dB in smooth copper. The 165.8 mm board carries about 60 mm of feed line, so the feeds cost 0.2 dB of the total.
- *Copper roughness.* Doubling the conductor loss ($K = 2$) would add 0.6 dB. The measurement suggests that it is lower.
- *Bandwidth.* The loss scales as 1/FBW: for the same Q_u, a 10 % filter would lose half as much as this 5 % one.
- *Laminate.* The loss scales with $\tan\delta$: a 10 % change in $\tan\delta$ moves the dielectric part by 0.1 dB. The band edges lose more than the centre (2.58 dB in Palace) because the group delay peaks there.

A 1.52 mm RO4350B board only reaches IL ≈ 1.3 dB, because Q_d still dominates. RT/duroid 5880 ($\tan\delta \approx 0.0009$) at 1.52 mm would bring it to about 0.5 dB.

Cross-solver check. CST Frequency Domain, run from the generated macro, agreed with Palace on fc (2.876 vs. 2.875 GHz) and IL (1.80 vs. 1.85 dB) on the untrimmed board. Getting there took ten debugging steps on the macro and the solver setup. Seven of them were actual fixes. We also built a coax-only control model, to separate port problems from filter problems. In Table VI the CST run gives 2.878 GHz, 180 MHz and 1.80 dB. Palace gives 2.876 GHz, 182 MHz and 1.70 dB on the fabricated geometry. The CST geometry is the untrimmed one. The time-domain solver was never made fully stable.

The connector is part of the design. Figs. 11 and 12 compare three launches on the same geometry: a bare microstrip feed, the CPWG launch without a connector, and the CPWG launch with the SMA. Without the connector only one reflection zero remains, and the worst in-band S11 is −6.3 dB (microstrip) or −5.1 dB (CPWG), against −11.1 dB with it. The filter was tuned with the connector's reactance in place. The bare CPWG port also shows a port-modelling trap. Its first two Palace modes belong to the ground pours (S21 ≈ −78 dB). The signal mode is mode 3.

Manufacturing. The same command-line tool writes the KiCad board, the gerbers, the drill files and a STEP solid. It also runs 34 automated file-level checks [38]. The checks include drawn-versus-meshed geometry and CST-versus-mesh via positions. JLCPCB stocks RO4350B 0.76 mm but offers only 1 oz outer copper. That leaves the finished thickness at about 0.84 mm at the pads. The SMA fork slot is 0.81 ± 0.1 mm, so the fit is marginal. It can be fixed by removing 15–20 µm of copper under the connector

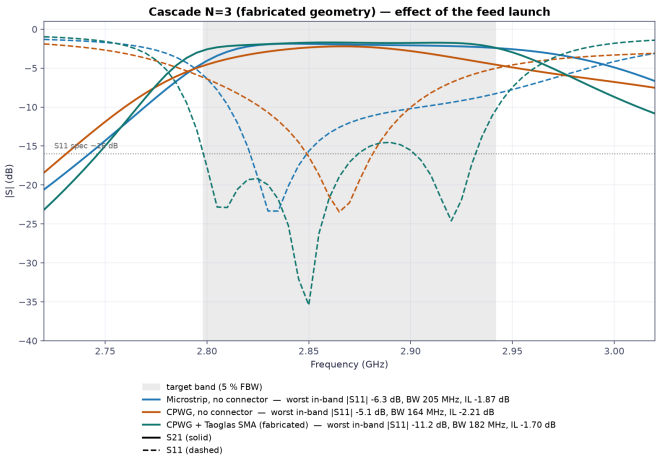


Fig. 11. Same geometry, three launches. Removing the SMA costs ≈5 dB of return loss because the design was tuned with it in place.

legs. In practice the author did this with a nail file until the connector seated in the slot. The gerbers carry the no-mask requirement, and the order notes state it. The boards were ordered from JLCPCB on Rogers RO4350B with an electroless nickel immersion gold (ENIG) finish. The order cost about USD 120, laminate surcharge included. We chose ENIG because the unmasked copper is exposed and ENIG is a durable barrier. We assumed that the plating, which sits on the air side of the copper, does not touch the cavity fields. The measurement cannot exclude it on the feed lines (Sec. VIII).

Measured results

Setup and calibration

The board (Fig. 13) was measured with a Keysight FieldFox N9917A handheld vector network analyzer (18 GHz). The sweep ran from 100 MHz to 5 GHz with 10 001 points, 10 kHz IF bandwidth and −15 dBm source power. We calibrated the instrument at the ends of the two test cables. The mode was full two-port SOLR (short-open-load-reciprocal), with a generic SMA kit. The short, open and load were connected on each port. Both cables end in SMA plugs, so they cannot mate with each other. The thru was therefore a female-female SMA adapter of unknown length and loss. SOLR is meant for exactly this case: the R stands for a reciprocal, unknown thru. We omitted isolation. The generic standards are not a characterised vendor kit. We set the instrument to the definitions of the 3.5 mm 85033D/E kit, the nearest it offers. The true offset delays, open capacitance and load return loss are therefore unknown. The plugs mate directly with the board's jacks, so the reference plane is the connector face, as in the simulations. We saved four states of the same board: connectors soldered, or seated against the board without solder, each with the instrument calibration applied and without it. We also saved the raw open, short, match and thru, at the same plane.

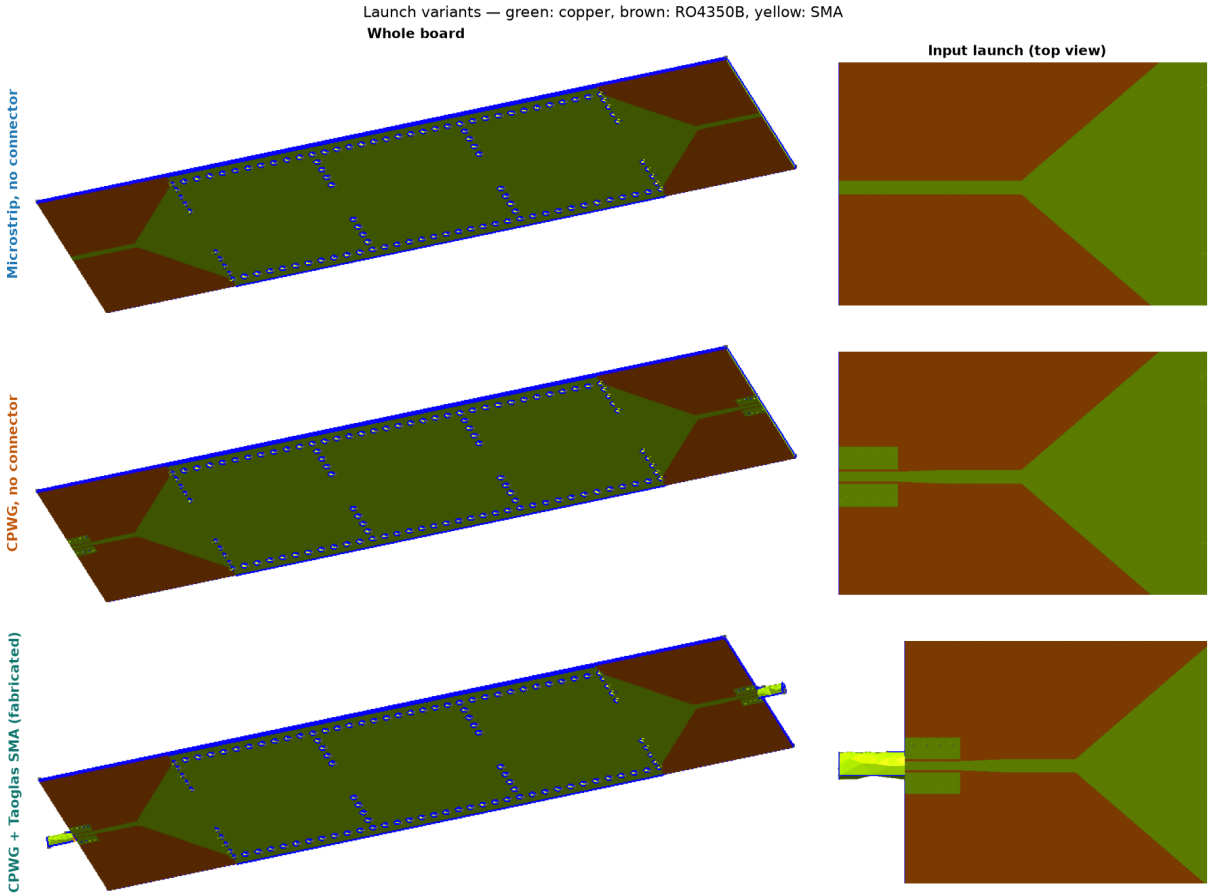


Fig. 12. Palace models of the three launches: whole board and input-launch close-up. Green: copper; brown: RO4350B; yellow: SMA connector.

Software recalibration. The raw standards allow a second, independent correction. We solved the eight-term error model in software with scikit-rf [39]. Each port gets a one-port short-open-load calibration. The unknown-thru step gives the transmission terms [40]. We applied the result to the two uncalibrated boards.

How it works, in short. The analyzer does not see the board directly. Between its receivers and each connector face sit cables, the switch and the instrument's own electronics. We model that path at each port as an *error box*, an unknown two-port. This is the eight-term model, with seven independent terms per frequency. To calibrate, we measure known standards and solve for the two boxes. To correct, we remove the boxes from the filter measurement. At each frequency there are three steps:

1. *One port at a time (SOL).* At port 1, the measured reflection Γ_m and the true reflection Γ are related by
$$\Gamma_m = e_{00} + \frac{e_{10}e_{01}\,\Gamma}{1 - e_{11}\Gamma},$$
where e_{00} is the directivity, e_{11} is the source match and $e_{10}e_{01}$ is the reflection tracking. Multiplying out the denominator makes the relation linear: $\Gamma_m = e_{00} + \Gamma\,\Gamma_m\,e_{11} - \Gamma\,\Delta_e$ with $\Delta_e = e_{00}e_{11} - e_{10}e_{01}$. The three standards have known Γ (short ≈ -1, open $\approx +1$, load ≈ 0). They give a 3×3 system. Port 2 gives e_{33}, e_{22} and $e_{23}e_{32}$ in the same way.
2. *Transmission from the unknown thru.* One term is left: the transmission tracking $e_{10}e_{32}$, which links the two boxes. Classic SOLT takes it from a zero-length thru. No such thru exists here, because the female-female adapter has an unknown length and loss. The unknown-thru method [40] uses only one fact: the adapter is reciprocal, $S_{21} = S_{12}$. With the boxes of step 1 known, that condition gives
$$(e_{10}e_{32})^2 = e_{10}e_{01}\;e_{23}e_{32}\;\frac{S_{21,m}}{S_{12,m}},$$
where $S_{21,m}$ and $S_{12,m}$ are the adapter's measured transmissions. The root has two solutions of opposite sign. At each frequency we keep the one that leaves the corrected adapter's phase closer to an ideal thru (within $\pm 90°$). The sign changes only the phase of S_{21}, not the magnitude we report.
3. *Correction.* With both boxes known, we convert each filter measurement to transfer parameters T, in which a cascade is a product: $T_m = T_A\,T_{\mathrm{DUT}}\,T_B$. So $T_{\mathrm{DUT}} = T_A^{-1}\,T_m\,T_B^{-1}$. We then convert back to S.

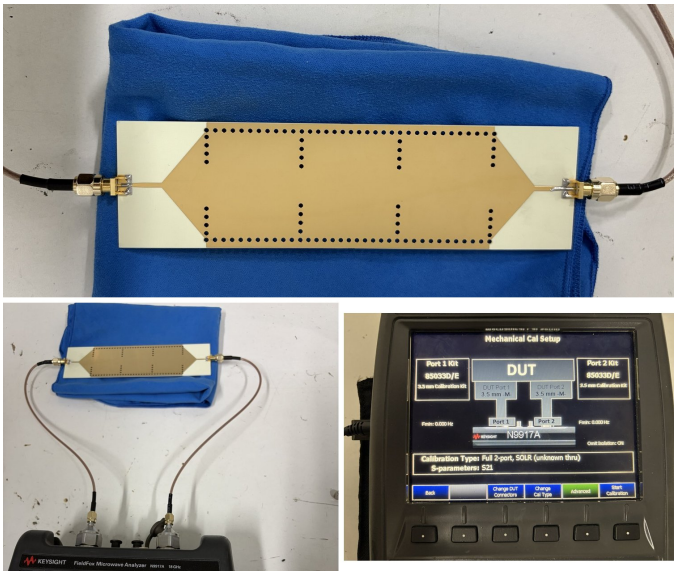


Fig. 13. Measurement setup. Top: the fabricated board with its connectors and test cables (no soldermask, ENIG finish). Bottom left: bench setup with the VNA. Bottom right: the instrument's calibration screen (3.5 mm 85033D/E definitions selected, full two-port SOLR with unknown thru).

We did not measure the instrument's switch terms. We assume that the switch is matched and that isolation is negligible. The open and short follow the definitions the instrument was set to. They use a model of the 85033D/E, with about 30 ps offset delay and 49 fF fringing capacitance. These are typical values, not values read from the instrument. The result agrees with the instrument-corrected files over 2.6–3.2 GHz (Fig. 14). The mean $|\Delta\Gamma|$ is 0.03 for S_{11} and 0.04 for S_{22}. For S_{21} the mean difference is 0.17 dB. If we use ideal $+1/-1/0$ standards instead of the kit model, the magnitudes change by less than 0.03 dB (S_{21}) and 0.09 dB (S_{11}). The standards only move the reference plane.

Comparison with TOSM. Classic TOSM (thru-open-short-match, Keysight's SOLT) uses the same standards. It is identical to our method in steps 1 and 3. Only step 2 changes. TOSM declares the thru ideal and of zero length ($S_{11} = S_{22} = 0$, $S_{21} = S_{12} = 1$). With that assumption, the thru measurement gives the transmission term directly:

$$S_{21,m} = \frac{e_{10}e_{32}}{1 - e_{11}\,e'_{22}} \;\Rightarrow\; e_{10}e_{32} = S_{21,m}\,(1 - e_{11}\,e'_{22}),$$

$$e'_{22} = \frac{S_{11,m} - e_{00}}{S_{11,m}\,e_{11} - \Delta_e},$$

where e'_{22} is the port-2 load match seen from port 1 through the thru. We find it by inverting the step-1 equation (12-term model; the reverse direction is analogous). With an unknown thru, $S_{21,m}$ and $S_{12,m}$ share the same denominator. Their ratio cancels it, together with the adapter's own transmission. This is why the unknown-thru method (UOSM, here SOLR) does not need to know the thru, and TOSM does. If the thru is not ideal, TOSM loads its loss, phase and mismatch into the error terms.

To size that discrepancy, we calibrated the same raw files both ways, with the same standard definitions (Table V). At 2.87 GHz, TOSM changes the transmission term by +0.33 dB and −3.4°. By construction, it leaves the corrected adapter at 0 dB and 0°. With the unknown thru, the adapter reads +0.27 dB and −3.1°. The instrument reads it at +0.18 dB and −2.2°. On the soldered board, TOSM raises the insertion loss from 2.04 to 2.39 dB, narrows the band by 2.3 MHz and moves f_c by 2 MHz. The worst in-band S_{11} barely changes, but the best one deepens by 4 dB. The reason is that in TOSM e'_{22} comes from the thru's S_{11}. Declaring the adapter's reflection zero (−22 dB after the unknown-thru correction) moves that reflection into e'_{22}. The TOSM loss matches the instrument's (2.38 dB) only by coincidence. The instrument did not treat the adapter as ideal, as its +0.18 dB reading shows. Over the rest of the band, TOSM is no closer to the instrument in transmission (mean ΔS_{21} +0.14 against +0.17 dB). It is further away in reflection (mean $|\Delta\Gamma|$ 0.041 against 0.030). The two methods differ by about 0.35 dB in insertion loss. This is of the order of the 0.3 dB uncertainty assigned to S_{21} below.

Both corrections assume the same standard definitions. Their agreement therefore checks the implementation and the saved raw files, not the standards. The offsets and the open's capacitance mostly move the reference plane. The test above bounds their effect on the magnitudes. The adapter itself shows this. Corrected with the 85033D/E definitions, it has less than 1 ps of delay. Corrected with ideal standards, it has −64 ps. The assumed definitions move the reference plane by about 32 ps per port, while the magnitudes change by less than 0.09 dB. The load is different. A general-purpose SMA load with a 25 dB return loss, which we did not measure, would leave a directivity error of 0.056. This is comparable to the dips we want to read. The check thru sets an empirical bound. After the instrument calibration, it returns +0.18 dB in S_{21} and −18.4 dB in S_{11} at worst over 2.6–3.2 GHz. The software calibration applied to the raw thru gives +0.26 dB and −17.6 dB. A passive thru cannot have gain, so both corrections are optimistic in S_{21}. We take the uncertainty of S_{21} as about 0.3 dB. Reflections below about −18 dB are not resolved. The worst in-band S_{11} is good to about ±2 dB.

Comparison with simulation

Table VI compares the specification, both simulators and the four corrected measurements, all processed by the same code (`siw measure`). Fig. 15 shows the curves.

How f_c and the bandwidth are read. One definition is used throughout the paper and for every source. On each curve we take the maximum of $|S_{21}|$ and draw a line 3 dB below it. The line crosses the curve on each side of the peak. We interpolate linearly between sweep points to

TABLE V
SOFTWARE CALIBRATION OF THE SAME RAW FILES WITH AN UNKNOWN THRU (THE REPORTED ONE) AND WITH TOSM, WHICH DECLARES THE FEMALE-FEMALE ADAPTER AN IDEAL ZERO-LENGTH THRU. SOLDERED BOARD; THE LAST COLUMN IS THE INSTRUMENT'S OWN CALIBRATION (SOLR, UNKNOWN THRU). THE ADAPTER ROW IS THE THRU ITSELF AFTER CORRECTION, AT 2.87 GHZ. MEAN DIFFERENCES OVER 2.6–3.2 GHZ.

	Unknown thru	**TOSM**	**Instrument**
$e_{10}e_{32}$ at 2.87 GHz	−12.04 dB ∠81.7°	−11.71 dB ∠78.3°	–
Corrected adapter S_{21}	+0.27 dB ∠−3.1°	0 dB ∠0° (by definition)	+0.18 dB ∠−2.2°
f_c (GHz)	2.8945	2.8926	2.8943
BW (MHz)	172.5	170.2	171.2
Peak IL (dB)	2.04	2.39	2.38
Best S_{11} (dB)	−18.8	−22.8	−16.8
Worst in-band S_{11} (dB)	−4.6	−4.5	−5.3
Mean ΔS_{21} vs. instrument (dB)	+0.17	+0.14	–
Mean $\lvert\Delta\Gamma\rvert$ S_{11} vs. instrument	0.030	0.041	–

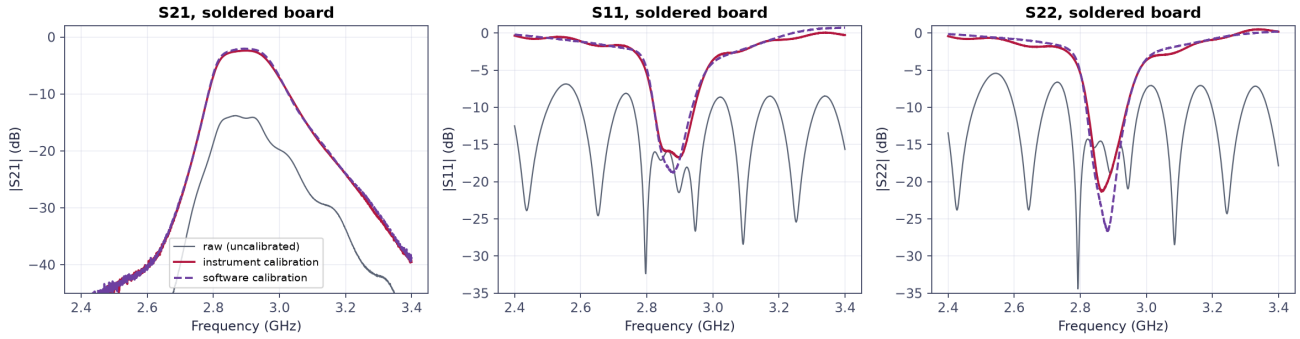


Fig. 14. Soldered board: raw measurement, instrument calibration and software calibration. The software result uses only the raw open, short, match and thru files.

find the two crossings. They are the band edges f_1 and f_2. The bandwidth is $f_2 - f_1$. The centre frequency is their midpoint, $f_c = (f_1 + f_2)/2$, not the frequency of the maximum. With ripple in the passband, the maximum sits on one of the poles and can be well off centre. In Palace the maximum is at 2.860 GHz and $f_c = 2.876$ GHz. On the soldered board the two coincide at 2.894 GHz. Only the crossings that bound the contiguous region around the peak count. A band split in two is therefore not reported as one wide band. Fig. 16 shows the construction on the curves.

Centre frequency and bandwidth. The measured band is centred at 2.894 GHz (soldered) and 2.898 GHz (unsoldered), against 2.876 GHz in Palace and 2.870 GHz in the specification. The +18 MHz (0.6 %) offset to the simulation corresponds to an effective permittivity about 1.3 % below the 3.66 used for design. This is well inside the spread between a laminate's design and process values. We did not test this. The −3 dB bandwidth is 171–174 MHz, 4–6 % below Palace.

Insertion loss. The peak S_{21} is −2.38 and −2.41 dB with the instrument calibration and −2.04 and −2.08 dB with the software one. So the loss is 2.0–2.4 dB, within the 0.3 dB uncertainty above. Palace gives 1.70 dB. The budget of Sec. VII expects 2.3 dB for smooth copper. The first version of the budget, written before the measurement, gave 2.1–2.3 dB. The measurement is therefore at or just below the expectation. Sec. IX turns the loss into a quality factor. A second route is the forward model of the fitted coupling matrix. It can read the measured loss as an effective unloaded Q. We kept the Palace-fitted couplings and varied only Q_u. The model gives 1.1 dB and three reflection zeros at $Q_u = 270$. It gives 1.7 dB and two zeros at 180, and 2.3 dB and one zero at 130 (Fig. 15b shows one). Read straight off the model, the measured 2.0–2.4 dB means Q_u of about 125–150. The matrix has no launch. Palace shows 0.4 dB more than the matrix at $Q_u = 229$, and removing that gives 150–185. We do not separate copper roughness, via-fence leakage, solder joints, the connector and the ENIG finish on the feed lines.

Return loss. The trace shows one broad minimum with a shoulder. It is −17 dB at best with the instrument calibration (−19 to −20 dB with the software one). The worst in-band S_{11} is −5 dB. The three simulated zeros are at −23 to −35 dB. This is below the −18 dB floor of the calibration check, so the measurement cannot confirm them one by one. The matrix model says that this much loss merges them into one. That is what the trace looks like. The 24 MHz offset from the specified band adds to the in-band figure. The ports differ slightly: the best S_{22} is −21 dB against −17 dB for S_{11} on the soldered board. This is within the calibration floor.

Soldering. Soldering the connectors changes the calibrated result by 3 MHz in centre frequency, 0.03 dB in insertion loss and 0.5 dB in worst in-band S_{11}. At 2.9 GHz the contact of the seated connector is adequate. The fork-slot

TABLE VI

FABRICATED CASCADE N=3 FILTER: SPECIFICATION, SIMULATION AND MEASUREMENT. EVERY COLUMN IS PROCESSED WITH THE SAME CODE: f_c IS THE MIDPOINT OF THE TWO CROSSINGS 3 DB BELOW THE PEAK S_{21}, AND THE BANDWIDTH THEIR DISTANCE. IN-BAND IS $2.870 \pm 2.5\%$ GHZ, I.E. 2.798–2.942 GHZ (THE SPECIFIED 5 % BAND). CST IS THE PRE-TRIM GEOMETRY. PALACE IN-BAND FIGURES ARE FOR THE FABRICATED GEOMETRY; ITS S_{21} AT 2.45/3.5 GHZ AND SPUR ARE FROM THE PRE-TRIM 0–8 GHZ SWEEP. THE SPUR IS THE LARGEST S_{21} BETWEEN 3.6 AND 5.0 GHZ. MEASURED COLUMNS: VNA = INSTRUMENT CALIBRATION, SW = SOFTWARE RECALIBRATION OF THE RAW FILES (SEC. VIII); S = CONNECTORS SOLDERED, U = UNSOLDERED, C = CALIBRATED. Q_L AND THE EFFECTIVE Q_u ARE DEFINED IN SEC. IX. THE SPECIFIED BANDWIDTH IS THE 0.1 DB RIPPLE BANDWIDTH; THE SPECIFICATION Q_L IS THAT OF THE IDEAL PROTOTYPE AT −3 DB (199 MHZ), NOT $1/\mathrm{FBW} = 20$.

	Spec.	**CST**	**Palace**	**VNA SC**	**VNA UC**	**SW S**	**SW U**
Centre frequency (GHz)	2.870	2.878	2.876	2.894	2.898	2.895	2.898
−3 dB bandwidth (MHz)	143.5	180	182	171	172	172	174
Fractional bandwidth (%)	5.0	6.2	6.3	5.9	5.9	6.0	6.0
Loaded $Q_L = f_c/\mathrm{BW}$	14.4	16.0	15.8	16.9	16.9	16.8	16.7
Effective Q_u (from IL)	–	173	179	138	136	159	155
Peak S_{21} (dB)	–	−1.80	−1.70	−2.38	−2.41	−2.04	−2.08
Worst in-band S_{21} (dB)	–	−3.76	−2.58	−7.14	−7.62	−6.79	−7.26
Worst in-band S_{11} (dB)	≤ -16	−6.9	−11.1	−5.3	−4.8	−4.6	−4.1
Worst in-band S_{22} (dB)	–	−6.9	–	−5.2	−4.5	−5.0	−4.3
Best S_{11} (dB @ GHz)	–	−17.3 @ 2.877	−35.4 @ 2.850	−16.8 @ 2.899	−17.1 @ 2.884	−18.8 @ 2.881	−20.1 @ 2.873
S_{11} zeros < -10 dB	–	1	3	1	1	1	1
S_{21} at 2.45 GHz (dB)	–	−60.7	−60.9	−45.6	−45.8	−44.9	−45.0
S_{21} at 3.5 GHz (dB)	–	−39.3	−39.5	−43.3	−43.5	−41.6	−42.3
Strongest spur (dB @ GHz)	–	−2.5 @ 4.63	−3.8 @ 4.61	−4.4 @ 4.68	−4.2 @ 4.68	−4.5 @ 4.68	−4.4 @ 4.68

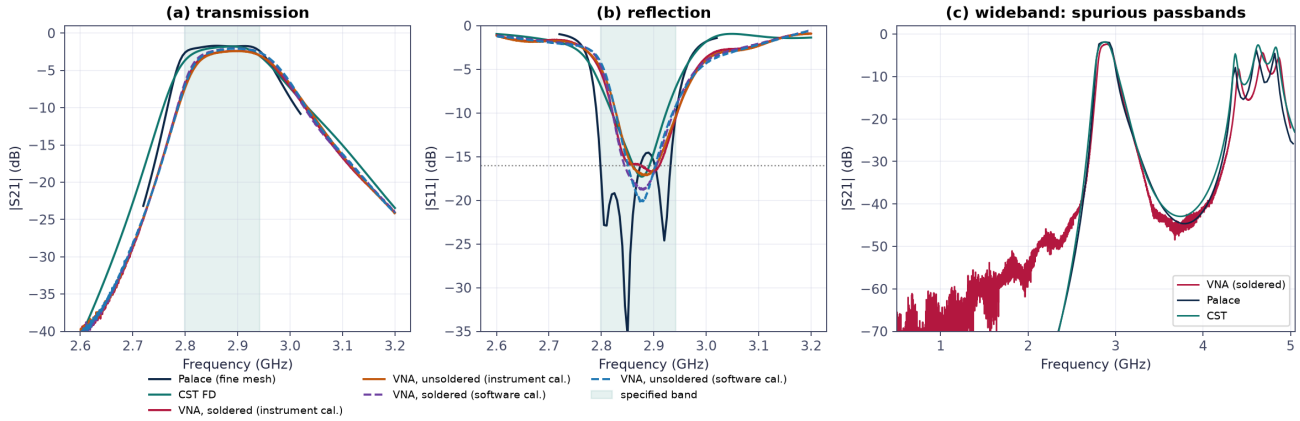


Fig. 15. Measured and simulated response. (a, b) The fabricated board (instrument and software calibration, soldered and unsoldered) against Palace and CST. (c) Wideband transmission: the spurious passbands at 4.4–4.9 GHz appear in the measurement at the simulated frequencies.

fit discussed in Sec. VII did not become a problem.

Out of band. The first spurious passband appears at 4.68 GHz and −4.4 dB, against 4.61 GHz and −3.8 dB in Palace and 4.63 GHz and −2.5 dB in CST. Above the passband the measurement stays within a few dB of Palace. At 3.7 GHz it is −45.6 dB against −44.2 dB. At 4.0 GHz it is −41.2 dB against −41.7 dB. At 3.5 GHz it is −43.0 dB against −39.5 dB. Below the passband the simulated rejection keeps falling (−61 dB at 2.45 GHz, −92 dB at 2.0 GHz). The measured rejection levels off at −44 to −48 dB between 2.2 and 2.5 GHz, and is −55 dB at 2.0 GHz. This is well above the instrument floor (below −70 dB). It points to coupling around the filter (open feed regions, cables) or leakage through the via fence. The models do not contain these paths.

TABLE VII

EFFECT OF THE LAUNCH ON THE SAME FILTER GEOMETRY. THE FIRST THREE ROWS ARE PALACE SIMULATIONS (THE PORTS OF THE TWO CONNECTOR-LESS VARIANTS SIT ON THE PLANAR LINE AND NORMALISE TO ITS OWN IMPEDANCE, SO THEY SHOW THE FILTER BEHIND AN IDEALLY MATCHED FEED). THE LAST TWO ROWS ARE THE FABRICATED BOARD (INSTRUMENT CALIBRATION). S_{21} IS THE PEAK AND S_{11} THE WORST VALUE IN THE SPECIFIED BAND.

Launch	f_c (GHz)	**BW (MHz)**	S_{21} (dB)	S_{11} (dB)
Microstrip, no conn.	2.897	205	−1.87	−6.3
CPWG, no conn.	2.875	164	−2.21	−5.1
CPWG + SMA, sim.	2.876	182	−1.70	−11.1
CPWG + SMA, sold.	2.894	171	−2.38	−5.3
CPWG + SMA, unsold.	2.898	172	−2.41	−4.8

Effect of the launch

Table VII adds the measured board to the launch simulations of Sec. VII.

The simulations without a connector use ports that normalise to the planar line. They cannot be compared

in absolute terms with a measured board, which has a connector and a real 50 Ω reference. They show that the connector's reactance is part of the tuning. The measurement shows that the board with its connector agrees with the connector simulation in centre frequency, bandwidth and spurious bands, and not in return loss.

All figures in this section are regenerated from the files in `measurements/2026-10-14_vna/` by `siw measure --paper`.

Quality factor: loaded and unloaded Q

Two quality factors summarise the response. The **loaded Q** of the filter, $Q_L = f_c/\mathrm{BW}_{-3\,\mathrm{dB}}$, says how narrow the passband is with the ports attached. We read it from $|S_{21}|$ as in Sec. VIII: the dotted line in Fig. 16 sits 3 dB below each curve's peak, and its ends are the band edges. The **unloaded Q**, Q_u, measures the loss of each resonator on its own. It sets the insertion loss (Eq. 5). The two Q rows of Table VI and Fig. 17 come from the same code (`siw measure`) for all four sources.

In the full filter the two play different roles:

- Q_L is set by the couplings (k and Q_e). It defines the bandwidth and barely depends on Q_u as long as $Q_u \gg Q_L$.
- Q_u does not change the bandwidth, but it sets the insertion loss. Eq. 5 can be written with the loaded Q of the ripple band, $Q_{L,\mathrm{ripple}} = 1/\mathrm{FBW}$, as

$$\mathrm{IL}_0 \approx 4.343 \sum_i g_i \, \frac{Q_{L,\mathrm{ripple}}}{Q_u} \ \mathrm{dB}.$$

  The loss is higher when the filter is narrower (higher Q_L). It is also higher when each resonator is lossier (lower Q_u).

Theoretical loaded Q. The specified 5 % is the 0.1 dB *ripple* bandwidth, 143.5 MHz. Its Q is 1/FBW = 20. At −3 dB, the ideal third-order Chebyshev prototype is $\cosh(\frac{1}{N}\operatorname{arccosh}\sqrt{(10^{0.3}-1)/(10^{0.01}-1)}) = 1.39$ times wider. That is 199 MHz, or $Q_L = 14.4$. This is the value to compare with, because every bandwidth in this paper is a −3 dB one. Palace gives 15.8, CST 16.0 and the board 16.9 (16.8 with the software calibration). Against this reference, the −3 dB band is not wider than designed. It is 9 % (Palace) to 14 % (board) narrower. This is consistent with the internal coupling that the matrix fit finds slightly below target ($k_{12} = 0.042$ in Palace and 0.044 on the board, against 0.046). Earlier comparisons against 143.5 MHz mix the two bandwidth definitions.

Effective Q_u. We solve Eq. 5 for Q_u with the measured peak loss. The FBW in that equation is the ripple one, so we divide the measured −3 dB FBW by 1.39. On the ideal lossy prototype, this procedure returns 270 for $Q_u = 270$ and 181 for 180. All the loss (mismatch, launch, radiation, connector) is charged to the resonators. The result is therefore an *effective* Q_u, a lower bound on the resonator's own. Palace gives 179 and CST 173. Both simulations have perfectly conducting walls, so their ceiling is the dielectric, $Q_d = 270$. The gap is the mismatch and launch terms of the loss budget (Table IV). The board gives 138 with the instrument calibration and 159 with the software one. Both lie inside the 130–185 range that the coupling-matrix forward model gave in Sec. VIII. The board sits at 0.77–0.89 of Palace. This is what copper, the ENIG finish and the real connector add. None of them is in the simulations. The smooth-copper reference, $1/Q_u = 1/Q_d + 1/Q_c$ with $Q_c \approx 540$, gives 180 for the resonator alone.

A second estimate, from the group delay. Eq. 5 needs the g-values and the bandwidth. A second estimate needs neither. A signal spends a time τ_0 in the filter. In that time, the stored energy decays by the factor $\exp(-\omega_0\tau_0/Q_u)$, which is $4.343\,\omega_0\tau_0/Q_u$ dB. Setting this equal to the dissipated loss gives

$$Q_u = \frac{4.343\,\omega_0\,\tau_0}{L_{\mathrm{diss}}}, \qquad L_{\mathrm{diss}} = -10\log_{10}\frac{|S_{21}|^2}{1-|S_{11}|^2}. \quad (9)$$

Here τ_0 is the group delay of S_{21} at the transmission peak (the phase slope over a 10 MHz aperture). L_{diss} is the power that is neither reflected nor transmitted. On the ideal lossy prototype, Eq. 9 returns 270 for $Q_u = 270$ and 180 for 180. Eq. 5 returns 270 to 277 and 181 to 187, within 4 %. Palace gives 222. The board gives 156 (instrument calibration) and 183 (software calibration). These are the diamonds in Fig. 17b, and Fig. 18 shows the delay.

The two estimators charge the feed lines differently. Eq. 5 puts all the loss on the resonators, so it is a lower bound on their Q_u. Eq. 9 also counts the delay of the feed lines and the launch as stored energy, so it reads higher. For Palace, whose cavities have exactly $Q_u = 270$, the two give 179 and 222. They agree on the main point: the board is 11 to 30 % below Palace. Adding smooth copper in series to the Palace value gives $1/(1/222 + 1/540) \approx 158$. The board lies there (156 to 183).

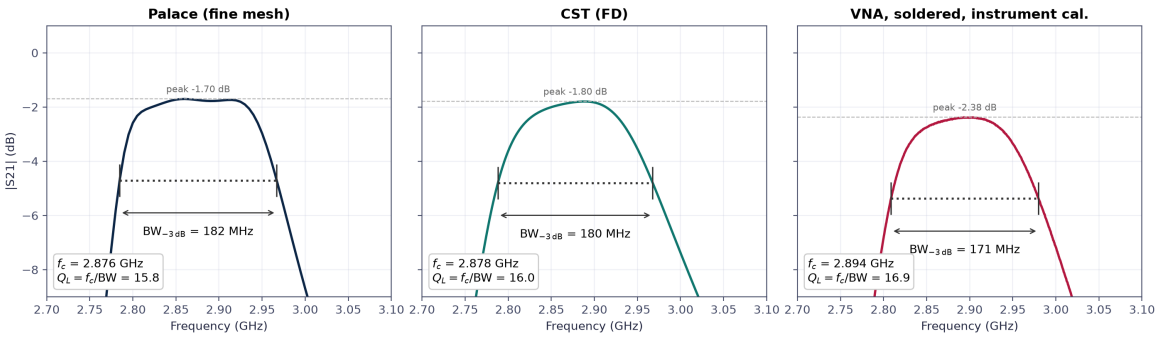


Fig. 16. Loaded Q read from $|S_{21}|$. The dotted line is 3 dB below each curve's peak and runs edge to edge; the arrow marks the bandwidth and the box gives $Q_L = f_c/\mathrm{BW}$.

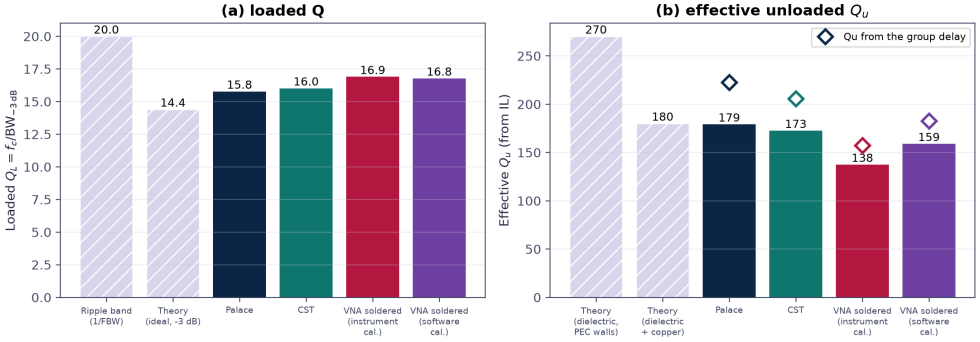


Fig. 17. Quality factors of the four sources against theory (hatched bars). (a) Loaded Q: the comparable value is the ideal prototype at −3 dB (14.4), not 1/FBW (20, ripple band). (b) Effective Q_u solved from the insertion loss, against the dielectric limit (270) and dielectric plus smooth copper (180). Diamonds: the same Q_u from the group delay, Eq. (9).

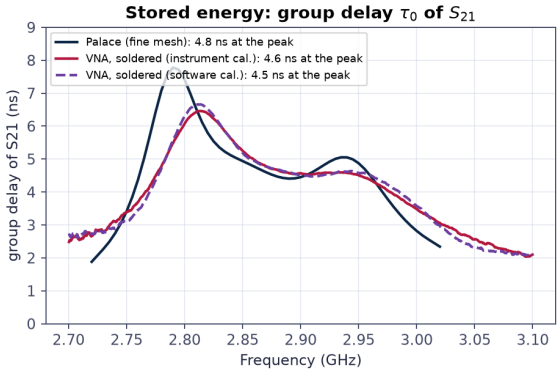


Fig. 18. Group delay of S_{21} for Palace and the soldered board. The value at the transmission peak is τ_0 in Eq. (9).

Discussion: gaps and room to improve

What was AI-driven, and what was not. The agent wrote about 12 k lines of design code. It launched and monitored the Palace eigenmode and driven solves, read solver logs, and localized most of the defects in Table III. It also produced the fabrication files. The engineer chose the specification, topology, laminate and vias, decided when to stop tuning, ran every CST check, and more than once rejected a confident but wrong explanation. The design physics that finally worked is classical. The acceleration came from automation and diagnosis, not from new electromagnetics. Stating this plainly is more useful than overclaiming.

Agent failure modes observed.

- Confident misdiagnoses: two wrong explanations of one stack failure were each falsified by a single re-run.
- APIs guessed by analogy: VBA properties that do not exist.
- Documentation drift: headline numbers kept in prose after the code had changed. The repository's status files had to be rewritten several times.
- Unreviewed edits from a parallel agent absorbed into a commit.
- Premature optimism about untested geometries.

Each was caught by a physics check, a solver error or a human question. This argues for keeping those checks in the loop.

Gaps.

- One board and one measurement session: no spread across boards, and the S21 scale is uncertain by about 0.3 dB (Sec. VIII).
- The calibration standards were a generic SMA short, open and load, not a characterised kit. The calibrated thru returns −18 dB, so reflection zeros deeper than that are not resolved, and the worst in-band return loss is good to about ±2 dB.

- The specification is not met: the board measures −5 dB worst in-band return loss against −16 dB (simulated: −11.1 dB). The −3 dB bandwidth is 171 MHz, against 199 MHz for the ideal prototype (the specified 143.5 MHz is the 0.1 dB ripple band, Sec. IX).
- S11 varied by about 5 dB across mesh and port models, while fc, bandwidth and IL moved by less than 10 MHz and 0.3 dB. The measurement then differed from Palace by +18 MHz in fc, 0.3–0.7 dB in IL and 6 dB in S11, so the model uncertainty was not conservative.
- Qe is calibrated with a dielectric-only Qu and is slightly over-estimated.
- Conductor loss is modelled only on exterior faces. Thin fabrication layers (mask, ENIG) could not be resolved.
- Spurious passbands start at 4.4 GHz (−4 dB, confirmed by the measurement) and need a low-pass filter in a real system.
- The measured loss is not apportioned between copper roughness, via-fence leakage, solder joints, connector and ENIG on the feed lines. The low-side rejection floor of about −46 dB (simulated: −61 dB) points to coupling around the filter that no model contains.
- There is no controlled comparison against an engineer working without the agent, and no accounting of cost. The session logs give tokens and active time (Table II) but not the engineer's own time.
- This is a single case study: one engineer, one model family, one filter topology. It shows what can happen, not how often.
- The taxonomy is self-reported by the same engineer–agent pair that made the mistakes. Git corroborates that each fix happened, but not the classification. It also lists only caught defects. The physics checks may still miss a comparable class of error.
- The code has no regression test suite. Its safety net is the physics sanity checks and the automated fabrication checks.

Improvements.

- A coordinated local correction of w_{ext}, l_{cav1} and w_{iris}, driven by the coupling-matrix fit, and a re-centring of the band. The +24 MHz offset from 2.870 GHz is of the order of 0.5–1 mm of cavity length. The sensitivity is 27 MHz/mm measured at the end cavities, and about 46 MHz/mm for the ideal cavity.
- Fitting the coupling matrix to the measurement needs a model of the launch. With the plain matrix the fit is ill-conditioned (Q_u between 57 and 85 for two nearly identical curves), so we read the loss through the forward model.
- Adaptive frequency sweeps and y-symmetric half-models, for about half the memory and 5–10× fewer solves, which would bring the cascade within GPU limits.
- A GPU with more memory.
- A lower-loss or thicker laminate.
- Space-mapping or surrogate models [2] trained on the calibration data.
- Repeating the measurement with a characterised SMA calibration kit, or with on-board thru-reflect-line standards, to resolve the zeros.
- Making the physics sanity checks a mandatory step of the agent loop rather than a habit.

The human part: how the text was written and what it means

How this text was written

The paper was written in a conversation, in many short rounds, and the engineer was never a passive reader. Each round had the same shape. The engineer asked for something or objected to something. The agent drafted or revised. Sub-agents that start without the conversation then reviewed the result and listed errors, and the engineer decided which findings to accept. Tables and figures of measured results are regenerated by `siw measure --paper`, so the numbers in the text can always be checked against code. Table VIII lists some of the engineer's prompts. They are paraphrased and corrected for grammar, so they show the process and are not literal quotes.

The agent also made mistakes in the text itself. It described the calibration standards as a vendor kit when they were a generic SMA kit (prompt 5). It wrote that the out-of-band rejection was a flat 45 dB, which is wrong: it is 44–48 dB between 2.2 and 2.5 GHz, where the simulation gives 61 dB. The second error was found in review. Both were corrected before release. This agrees with the failure modes of Sec. X: the agent is confident, and the checks have to come from outside.

What the engineer did that the agent did not

The physics that finally worked is classical, and most of the software was the agent's. It is still worth listing what the agent did not do, because a reader would otherwise credit it to the agent.

- *The idea of a coplanar-waveguide (CPWG) feed.* It came from a class that the author attended, not from the agent.
- *The physical work.* The SMA connector did not fit the board until the author filed the copper under its legs with a nail file (Sec. VII). No simulation could do that.
- *The measurement.* The author made sure that the VNA was used calibrated, and that the calibration was checked with a thru.
- *Reading.* The author checked the papers, the references and whether the equations looked right. Most of this was needed in the prototype period. After that the process went smoothly.
- *Free range.* The author also let the agent try other geometries that use other propagation modes. It was

TABLE VIII
EXAMPLES OF THE ENGINEER'S PROMPTS (PARAPHRASED) AND WHAT EACH ONE CHANGED.

#	Prompt (paraphrased)	What changed
1	"Reorganize the information we have, remove what is irrelevant, and write a paper called *AI-Driven RF Design*."	Repository cleanup (deleted files stay in git tags) and the first full draft.
2	"Say that we started with openEMS and why it did not work for us, then moved to Palace. Mention the GPU with remote control, and the Python scripts for the optimization. We ended up running on the CPU. Recognize the gaps, and reference every paper we used."	Phase 1 and 2 sections, the compute paragraph, the gaps list and the bibliography.
3	"I would prefer a venue that is not very restrictive in peer review. I do not mind if it is rejected. Check that LLM-written papers are accepted."	The arXiv and IEEE Access build targets, and a plain statement of LLM authorship.
4	"Run several reviewers on the paper, and analyze extremely carefully whether it is safe to make the repository public."	Corrections to claims in the text, and an audit of the git history and metadata before release.
5	"The calibration kit was not a vendor kit. It was a generic SMA short, open and load."	Text, README and code documentation corrected, and the -18 dB floor caveat added.
6	"Explain how the pipeline works: the equations that are needed, the traditional steps, and how the tool solves each one. Explain the optimizer too, since we wrote it around Palace. If needed, show the equations as display equations."	Table I, the optimizer paragraph and the display equations.
7	"Should we add today's commits? The paper is old. Some passages are hard to read, so simplify the wording. I do not know if simple technical English is acceptable in a paper, but I want it easy to follow."	Updated history statistics and timeline, and a plain-language editing pass.
8	"Is it easy to get the Q of the filter from the plots? Is the insertion loss an expected number? We got 1.7 dB in the simulation. Show how the real length and the copper change it, with an easy calculation."	Sec. VII loss budget and Sec. IX.
9	"I do not understand why we need Q_u or what it represents. Can we get it from Palace instead of CST?"	A plain explanation of Q_u, and the raw Palace run stored with its phase, so that the Palace value could be computed.

a blind search: without a physical idea of why a mode should help, the agent only explored (Sec. VI).

The simulation and optimization loop was the opposite. It is raw computation. It was easy to implement and to improve with the idea of an agentic loop: the agent runs a solve, reads the result, changes the model and runs again, and no human has to be in that loop.

Reflections and concerns

These are the author's views, offered with the evidence of one project.

Knowledge is no longer the scarce thing. The author's view is that AI has solved the problem of scarce knowledge. It reduces the time, and if it is guided correctly it does a better job than the author would have done alone. The coupled-resonator theory, the solver input format and the fabrication file formats were all things the agent knew or found in minutes. What it did not do was choose the problem, bring an idea from outside the text, or touch the board.

What is scarce now is money and the willingness to try new things. A simulation costs nothing but time, so the agent could try as many as it liked. The board cost about USD 120 and gave one measurement, and it was the only physical iteration of the project. The loop that the agent can run alone stops where the physical world begins. The second shortage is willingness: the agent tries whatever it is pointed at, and deciding to point it at an unfamiliar solver, a new topology or a new kind of board is still a human choice.

The gap between doing and knowing. What one can do with an agent grows faster than what one understands. In this project the gap was real. A working filter exists whose code the author did not write line by line, and Table III shows that the numbers can look right and be wrong. The defence was to check the result against physics, which is knowledge, not production.

Where the value goes. If production becomes instant, the value does not disappear. It moves to judgment and taste: choosing which thing to do, and seeing when the answer is wrong. In this project that meant the specification, the topology, the stopping decision and the doubts that a confident explanation did not remove.

Accountability. The agent does not want anything and is not answerable for its results. The author is, and says so in the acknowledgment [41].

CONCLUSION

An LLM coding agent took an SIW bandpass filter from specification to fabrication files, with one engineer supervising. The board was then manufactured and measured. Two things decided the design outcome. The first was to replace black-box search over a discontinuous objective with calibration of smooth observables on small structures. The second was to catch a long series of silent defects with physics checks. The measurement confirms the simulations where the physics is simple. The spurious bands fall at the

predicted frequencies, and the centre frequency is within 1 % of the specification. The measurement also shows what the simulations did not capture: 0.3 to 0.7 dB more loss than Palace. This is about the 0.6 dB of copper loss that the perfect-conductor walls leave out (Sec. VII). In the matrix model this loss merges the three reflection zeros into one. So the worst in-band return loss is 5 dB, not the simulated 11 dB and not the specified 16 dB. The generic calibration standards limit the reading of the zeros themselves (floor about −18 dB), not the 5 dB. The independent software recalibration agrees with the instrument's on average within 0.2 dB in S_{21} (up to 1 dB on the steep band edges). So this conclusion does not rest on one calibration. The code, data and scripts that produced every number are in the repository. `siw measure` reproduces the measured columns of Table VI. The agent made the computation cheap. The idea of the feed, the connector, the calibration and the decisions on what to trust stayed with the engineer (Sec. XI).

Acknowledgment

The design software, simulations, figures and a first draft of this text were produced with Claude Code (Anthropic) under the author's direction. The models were Claude Sonnet 5 and 5.5, Opus 5 and 5.5, and Fable 5. Sec. III describes this work. The author reviewed all results and text and takes full responsibility for them [41]. The complete history and data are available in the project repository.

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